Random Walks

Module 7 · Advanced Network Science
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(school logo, bottom left, as in the CCS deck)

#1 title

Where is the walker after 100 steps?

A man walks 1 yard, turns in a random direction, and walks 1 yard again. He repeats this 100 times.

Pearson asked this in *Nature* in 1905.

FIGURE

The first lines of Karl Pearson's letter to Nature, 27 July 1905 (facsimile or a re-typeset quotation, with credit). Under it: a pub, a lamp post, no answer yet

Far from the start, or near it? Tell your neighbour.

2

#2 hook (story): Pearson's question, 1905

Most likely near the start

FIGURE

LEFT: the paths of 5 walkers of 100 steps from the same lamp post (hand-drawn paths in pale colours, the pub in the middle). RIGHT: the end points of 10,000 walkers as a cloud of points centred on the pub; a grey ring at distance about 10 yards with the call-out 'typical distance after 100 steps'.

Rayleigh replied the next week: he had solved it in 1880, for waves

3

#3 answer: Pearson's walker (the density peaks at the s

Random walks appear in many fields

FIGURE

THREE PANELS in a row, each a picture with a one-line caption: (1) pollen grains in water, the zig-zag trace of one grain (Perrin's drawing, public domain): 'pollen, explained by Einstein in 1905'; (2) a mosquito: 'mosquitoes and malaria, the question behind Pearson's letter'; (3) a hand-drawn web graph with a surfer: 'the web, 1998: a random surfer ranks pages'. A grey arrow pointing right with the aside: 'and on a network?'

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#4 hook: the same walk in three fields

What does a random walk tell us about a network?

5

#5 spine 1: the question

One step: move to a random neighbour

FIGURE

Small 7-node network, hand-drawn. A walker (ring) stands on a node with 4 neighbours. Each of the 4 edges is labelled 1/4 (an amount, not a share). A grey aside: 'no memory of where it came from'.

6

#6 mechanism: the rule

LIVE TOOL

LIVE: random walk simulator, karate club, exactly as used last term (assets/vis/random-walks/index.html, no edits). The lecturer clicks Walk x20, then Run 5,000 steps.

7

#7 live: the simulator (karate club)

Short walk vs long walk

20 steps: what do you learn?

5,000 steps: what do you learn?

FIGURE

Two small screenshots of the tool: after 20 steps, after 5,000 steps (no answer marks yet)

Take a minute. Tell your neighbour.

8

#8 question to the room (Q1)

Short: neighbourhood. Long: degree.

FIGURE

LEFT: the tool after 20 steps, nodes shaded by visits: dark nodes cluster around the start node. RIGHT: after 5,000 steps: dark nodes are the hubs (33 and 0). Grey ring call-outs: 'the start', 'hubs'.

9

#9 answer to Q1

LIVE TOOL

LIVE: same tool, ?dataFile=two-cliques.json: two 5-cliques joined by a single edge.

10

#10 live: the simulator (two communities)

Two communities joined by one edge

How many steps until the walker crosses to the other community?

A: about 3

B: about 20

C: about 200

FIGURE

The 10-node two-community network, the walker in community 1, the single edge between the communities marked with a grey arrow

Take a guess.

11

#11 question to the room (Q2)

OPTIONAL

A short walk stays in its community

FIGURE

Same network. The walker's path drawn as a brush stroke, coloured by community (community 1 cool, community 2 hot); a time strip under it shows long runs of one colour, about 21 steps each. Grey call-out: 'about 21 steps per visit'.

12

#12 answer to Q2

Short walks show communities. Long walks show degree. Why?

13

#13 spine 2: the mystery

Pen & Paper, Part 1

10 minutes · stop at the line on the sheet

FIGURE

QR code (large) to the simulator, plus the short link

14

#14 Pen & Paper Part 1: opener

Part 1: tasks

1. Sketch: correlation with degree over time
2. Run 20,000 steps. Read the most and least visited node.
3. Compute degree ÷ share, for both

FIGURE

Large timer (10:00) and the QR code

15

#15 Pen & Paper Part 1: instructions

Correlation rises with the steps

FIGURE

Line plot. x: steps (log scale, 10 to 20,000). y: correlation between visit share and degree. Dots at 0.37, 0.53, 0.76, 0.89, 0.96, 0.99, 1.00 (mean over all 34 starts), a pale band for min to max. Grey call-out: 'at 20 steps, it depends on where you start'.

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#16 debrief Q1 (the sketch)

Degree ÷ share is constant

Karate club: 34 nodes, 78 edges, so 2m = 156

FIGURE

Scatter: x = degree, y = share of visits after 20,000 steps, 34 dots on a straight line through the origin. Node 33 (degree 17, share 0.109) and node 11 (degree 1, share 0.0064) circled, hot. Grey call-outs: '17 ÷ 0.109 = 156', '1 ÷ 0.0064 = 156'.

Share of time = degree ÷ 2m

17

#17 debrief Q2 (degree ÷ share)

Why does a long walk not depend on the start?

18

#18 spine 3: part question

The walk is defined by one matrix

FIGURE

LEFT (small): the 12-node 'two communities' network (nodes 0-5 and 6-11, two bridge edges 2-8 and 5-11). RIGHT: its transition matrix P as a heat map, 12 x 12, two dense blocks on the diagonal, thin entries off the blocks. Grey call-out on one row: 'row i = the odds of leaving i, sums to 1'.

19

#19 P (the 'two communities' network, the motif of the

One step is a multiplication by P

$$x(t+1) = x(t)\,P$$

FIGURE

Matrix picture in the M06 style: a wide bar (distribution x(t), 12 cells, shaded) times the square P = a wide bar (x(t+1)). Small network beside it with the walker's probability mass drawn as node shading. Grey call-out: 'x is a row vector: row x column'.

After t steps: $x(t) = x(0)\,P^t$

20

#20 x P

build 1/3

Rows of P^t after t steps

FIGURE

Heat map of P^1 (12 x 12), same crop and position as the previous step: two dense diagonal blocks. Callout: 't = 1: stays in its community'

21

#21 P^t as heat maps (1/3)

build 2/3

Rows of P^t after t steps

FIGURE

Heat map of P^5 (12 x 12), same crop and position as the previous step: the blocks blur. Callout: 't = 5: starting to leak'

22

#22 P^t as heat maps (2/3)

build 3/3

Rows of P^t after t steps

FIGURE

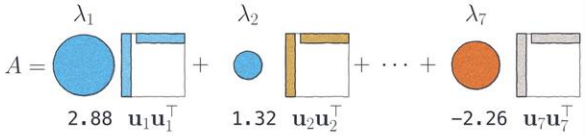
Heat map of P^40 (12 x 12), same crop and position as the previous step: every row identical. Callout: 't = 40: all rows = pi'

Every row becomes the same: π

23

#23 P^t as heat maps (3/3)

Recall: eigendecomposition



24

#24 recap: eigendecomposition (M06, light touch)

P has an eigendecomposition too

$P = \sum_k \lambda_k \, \phi_k \, \psi_k^\top$

FIGURE

Same pictorial grammar as M06: circle (eigenvalue, size = |lambda|), tall bar phi_k (right eigenvector), wide bar psi_k^T (left eigenvector); k = 1 blue, k = 2 gold, the rest pale; shown as terms 1, 2, ..., 12. A grey aside: 'for P, the tall and wide bars are two different vectors (each is the other rescaled by degree)'.

P^t changes only the eigenvalues

25

#25 P as a sum of circles and bars

build 1/3

The eigenvalues shrink as t grows

FIGURE

P^1 drawn as a sum of the same terms (same canvas as the previous step); circle areas = lambda^1: circles at 1, 0.89, 0, 0, 0, -0.09, -0.2, -0.2, -0.27, -0.33, -0.38, -0.42. First circle always full size, blue. Callout: 'one step'

26

#26 P^t as circles (1/3)

build 2/3

The eigenvalues shrink as t grows

FIGURE

P^5 drawn as a sum of the same terms (same canvas as the previous step); circle areas = lambda^5: circle 2 at 0.55, the rest almost gone. First circle always full size, blue. Callout: 'lambda_2^5 = 0.55'

27

#27 P^t as circles (2/3)

build 3/3

The eigenvalues shrink as t grows

FIGURE

P^40 drawn as a sum of the same terms (same canvas as the previous step); circle areas = lambda^40: circle 2 at 0.01; circle 1 untouched. First circle always full size, blue. Callout: 'only the circle at 1 is left: every row is pi'

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#28 P^t as circles (3/3)

Which term stays the longest?

The 12 eigenvalues of P.

After many steps, which term is still there?

FIGURE

The 12 eigenvalue circles in a row, ordered by value; no mark yet

Take 20 seconds.

29

#29 Q3 (which term stays the longest)

The second term stays the longest

$\lambda_2 = 0.89$

0.89^t: 0.55 at $t = 5$, 0.31 at $t = 10$, 0.01 at $t = 40$

FIGURE

Same 12 circles, now drawn at t = 10 sizes; circle 2 hot, the rest pale

30

#30 A3

The distance to π decreases

FIGURE

Five small bar charts in a row: the walker's distribution x(t) on the 12 nodes for t = 0, 1, 3, 10, 40, starting at node 0. A flat reference line pi on each chart (pi = degree / 60). Under each: 'distance to pi' as a number (1-norm), shrinking to about 0.

31

#31 distance to pi (small multiples)

build 1/3

Mixing time

FIGURE

Curves: distance to pi against t, one thin grey line per starting node (12), and a black line for the worst start. Callout: 'where you start matters at first'.

32

#32 mixing time diagram (1/3)

build 2/3

Mixing time

FIGURE

Same plot plus a horizontal line at epsilon. Callout: 'how close is close enough?'

33

#33 mixing time diagram (2/3)

build 3/3

Mixing time

$t_{\text{mix}} = \min\{\, t : \max_{x(0)} \|x(t) - \pi\|_1 \leq \epsilon \,\}$

FIGURE

Same plot; the point where the worst-start line crosses epsilon marked hot, with a vertical line down to the axis labelled 'mixing time'.

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#34 mixing time diagram (3/3)

OPTIONAL

Q: mixing time of a 10-clique?

10 nodes, every pair connected (45 edges)

FIGURE

A 10-clique, hand-drawn, nodes in a ring; the walker on node 0

Take a guess: 1 step, 5 steps, 50 steps.

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#35 Q4 (clique)

OPTIONAL

A: about 1 step

10 nodes, every pair connected (45 edges)

FIGURE

Same clique, same position. Distance-to-pi curve as an inset: below epsilon at t = 1

$\lambda_2 = -1/9$: mixing takes **about 1 step**

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#36 A4

Q: mixing time with one bridge?

Two 5-cliques joined by one edge (21 edges)

FIGURE

The two-clique network (the tool's two-cliques.json), the walker in community 1

The same, 5 times longer, or 15 times longer?

37

#37 Q5 (two cliques)

A: about 15 times longer

Two 5-cliques joined by one edge (21 edges)

FIGURE

Same network. Inset: distance-to-pi curve with the clique's curve in pale for comparison

$\lambda_2 = 0.93$: mixing takes about **14 steps**

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#38 A5

The mixing time follows from λ_2

$$\tau = \frac{1}{1 - \lambda_2}$$

FIGURE

A horizontal gauge for lambda_2 from -0.2 to 1 with markers: 10-clique (-0.11, tau 0.9), karate club (0.87, tau 7.6), two communities (0.89, tau 9), two 5-cliques + bridge (0.93, tau 14), ring of 1,000 (0.99998, tau 50,000). Past 0.99 the gauge is shaded hot.

Small gap, long mixing time

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#39 spectral gap (gauge)

OPTIONAL

Mixing time is a few relaxation times

FIGURE

The formula t_mix < tau log(1/(epsilon pi_min)) on the left; on the right the 12-node distance curve with tau marked and 2 tau, 3 tau as ticks.

40

#40 (optional) the bound

One eigenvalue is always 1. None is larger.

41

#41 spine 4: the fact

Why 1? Every row of P sums to 1

FIGURE

Picture: the square P times a tall bar of all ones (cells '1') = the same tall bar. One row of P highlighted; its cells sum to the first cell of the result (= 1). Callout: 'P 1 = 1'.

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#42 lambda = 1 exists (P 1 = 1)

No eigenvalue is larger than 1

FIGURE

Two frames on a 7-node network: numbers on the nodes (largest = 9); one step replaces every number by the average of its neighbours; the largest number can only stay or fall. Frame 2 is the same drawing with the new numbers; the largest is now 6.5.

43

#43 no eigenvalue above 1

The eigenvalue 1 gives π

The stationary distribution: $P^t \to \mathbf{1}\,\pi$

FIGURE

The first term alone: a tall bar of ones times a wide bar pi = a matrix whose every row is pi (the last heat map of the previous build, at the same position). Beside it the simulator's bars with the orange k/2m ticks (thumbnail).

$\pi_i = k_i/2m$: proportional to degree

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#44 lambda = 1 is pi (ties to the tool)

Pen & Paper, Parts 2 and 3

20 minutes

45

#45 Pen & Paper Parts 2-3: opener

Parts 2 and 3: tasks

2. Four nodes: write $x(t)$ with the eigenvalues, check $t = 2$
3. Twelve nodes: run one cell, plot the second eigenvector

FIGURE

Left: the 4-node "diamond" network of Part 2. Right: QR code to the one-cell notebook of Part 3

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#46 Pen & Paper Parts 2-3: instructions

$x(t) = \pi$ + decaying terms

Four nodes; eigenvalues $1,\ 0,\ -\frac{1}{3},\ -\frac{2}{3}$

FIGURE

The 4-node diamond network (degrees 2, 2, 3, 3) on the left. Right: bar charts of x(t) for t = 0, 1, 2, 5, 10 converging to pi = (0.2, 0.2, 0.3, 0.3). Under them the expansion: pi - 0.5 (0)^t psi_2 + 0 psi_3 + 0.3 (-2/3)^t psi_4, with the t = 2 line checked against x(2) = (1/3, 1/3, 1/6, 1/6).

Only the term with $\lambda = 1$ remains

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#47 debrief Part 2

The second eigenvector separates the communities

FIGURE

Point plot: x = node index 0 to 11, y = second eigenvector of P (scaled to max 1). Nodes 0-5 on top (about +1; the bridge nodes 2 and 5 at about +0.74), nodes 6-11 on the bottom (about -1; 8 and 11 at -0.74). Community 1 cool, community 2 hot. A dashed line at 0. Callout: 'the sign gives the community'.

48

#48 debrief Part 3 (second eigenvector)

Early on, the community signal dominates

Walker starts at node 0

FIGURE

Line plot: x = steps (0 to 40), y = probability of being in community 1: 1.00, 1.00, 0.92, 0.88, 0.80 (t = 5), 0.67 (t = 10), 0.55 (t = 20), 0.505 (t = 40). A dashed curve 0.5 + 0.5 x 0.889^t over it. Callout at t about 3: 'the other eleven circles are gone'.

The community signal decays like λ_2^t. Degree remains.

#49 debrief Part 3 (what dominates early)

FIGURE

Summary, three rows, each with a thumbnail of its figure: (1) the eigenvalue 1 = degree (thumb: scatter, slide 17). (2) The second eigenvalue = the communities, and its size determines the mixing time (thumb: gauge). (3) Short walks see communities; long walks see degree (thumb: the community-probability curve).

50

#50 summary (claims with thumbnails)

Next: how long does it take to reach a node?

FIGURE

The opening picture again: the karate club, nodes shaded by long-run visits, with two nodes marked A and B

51

#51 close: back to the opening picture, then a question